

Ashitha Paramesha Gowda

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Summary

Backend-focused full stack engineer with 4+ years of experience building distributed systems using Node.js, Python, PostgreSQL, and AWS. I operate with an AI-first, agentic mindset, rapidly experimenting with LLMs and autonomous workflows to design scalable, privacy-aware, real-time data systems.

Education

Johns Hopkins University, Maryland, USA Aug 2023 – May 2025

Master of Science in Engineering, Computer Science (3.5/4)

Coursework: Foundations of AI, Software System Design, Software Engineering, Computer Architecture, Computer Vision, Applied Cryptography, Security and Privacy in Computing, Future Networks (5G, SDNs).

Vidyavardhaka College of Engineering, Karnataka, India Aug 2017 – May 2021

Bachelor of Engineering, Information Science (3.8/4)

Experience

Founder, Full Stack Developer Jan 2021 – Present

cnotes.in, Bengaluru (Startup - **B2B SaaS** firm)

- Built & scaled a CRM & billing platform (**PHP, MySQL**) serving **30K+ subscribers**, replacing manual workflows with real-time forecasts (0.4–5% error) & driving **95% retention**.

Reserach Assistant - Gen AI Systems Mar 2026 – Present

Weill Cornell Medicine, New York

- Built a self-correcting **multi-agent pipeline** using **Google ADK, Gemini 2.5 Flash** extracting 150+ variables from 300+ Medicaid PDFs at 90%+ accuracy, processing 300+ waivers fully autonomously at scale.
- Implemented **model-agnostic self-correcting** critic agent with RAG-based inference, and persistent ChromaDB correction memory recovering **30+ percentage of field accuracy** via semantic LLM judgment.

Founder, Software Engineer Mar 2025 – July 2026

Dear: AI-driven Voice Assistant for Memory Loss

- Architected a **Dockerized, service-oriented backend** (**Node.js, FastAPI, ChromaDB, Redis**) powering real-time audio **transcription agent, memory parsing agent**, and retrieval-augmented dialogue generation agent.
- Built a full **React/Next.js UI** for Dear, including audio upload flows, memory visualization screens, agent interactions, and real-time voice playback, enabling E2E testing and rapid iteration of GenAI features.

Software Engineer (Research) – GenAI Systems Mar 2025 – Feb 2026

Johns Hopkins University, Baltimore, MD

- Built Python-based **GenAI extraction pipeline** with **70-field schema**, curating 3K+ labeled data points across 200+ Medicaid PDFs, developing reusable layouts for high-accuracy extraction and replacing 15+ manual workflows.
- Fine-tuned **Gemini 2.0** on the custom dataset & deployed it via **Vertex AI** with **95% accuracy**, boosting data coverage from **~15 fields to 70+** & saving **100+ analyst hours**.

Software Engineer II May 2023 – July 2024

JKTech, Bengaluru (Mid-sized - **B2B SaaS** firm, **Project:** Inveniam.io – data platform for private real estate equity)

- Built an event-driven sync system using **Node.js (NestJS), TypeScript, & PostgreSQL** to track & sync file meta-data & cloud pointers; implemented **hash-based change detection** & recorded updates on-chain for **blockchain-based file verification**, reducing sync lag by 50%.
- Leveraged **EventStore** to replay events & trace production issues; resolved **80% of support tickets**, including **30% classified as critical**, by enabling version-aware debugging of folder metadata & sync flows.
- Engineered **Angular 12** SPA performance improvements including lazy loading, caching, and cursor-based pagination, with full backward compatibility, reducing unnecessary data fetches across the Inveniam.io platform.

Software Engineer I Jan 2022 – Apr 2023

Juego Studios, Bengaluru (Startup - **B2B2C** gaming SaaS firm)

- Architected a **real-time voice/video communication plugin for the NJS2 engine** using **WebRTC** with **Socket.io**; supported **1:1 & 1:N chat** with **400+ concurrent users**, reducing dev effort by **80%**.

Skills

AI/ML: Gemini, Vertex AI, OpenAI APIs, TensorFlow, PyTorch, scikit-learn, Hugging Face, AssemblyAI; LLM pipelines, JSONL, Computer Vision, LangChain, RAG pipelines, LLM fine-tuning, prompt engineering

Languages & Frameworks: JavaScript, TypeScript, Python, SQL, Java; Node.js, NestJS, Express, Flask, React

Databases & Infra: PostgreSQL, MySQL, MongoDB; AWS (Lambda, CloudFormation), GCP, Docker, Kubernetes, Jenkins, GitHub Actions

Architecture & Tools: REST, GraphQL, WebSockets, OAuth2, JWT, Kafka, RabbitMQ, Microservices, TDD, DDD, Jira